

Machine learning approaches for prediction of fine-grained soils liquefaction

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ABSTRACT

Since soil liquefaction is a dimension that increases the amount and severity of losses in an earthquake, it is vital to estimate the liquefaction potential accurately. Traditionally, several analytical inferences were made for the prediction of soil liquefaction. However, it is necessary to use machine learning methods to establish nonlinear relationships of soil physical characteristics and develop an accurate classification model. In this study, the applicability of seven different machine learning algorithms; decision trees, logistic regression, support vector machines, k-nearest neighbors, stochastic gradient descent, random forest, and artificial neural network, were investigated on a data set obtained from field experiments (Standard Penetration Test) on soils in Adapazari region after the 1999 earthquake. Performance metrics such as accuracy, recall, precision, F1 score, and receiver operating characteristic evaluated algorithms. As a result of experimental studies, the decision tree algorithm performed best on the dataset, with an overall accuracy of 90%. The decision tree model provides an easy and effective tool for evaluating ground liquefaction potential to decision-makers. As a result of the decision tree study, it was observed that the mean grain size (D_{50}) soil feature has the most significant effect on the liquefaction potential.

1. Introduction

In geotechnical engineering, the behavior of the soil under static loads is constantly examined, and the behavior of the soil under the influence of dynamic loads, mainly caused by earthquakes, is also discussed. The adverse effects caused by earthquakes in alluvial environments are much more than in other environments. This is called the amplification effect, reduction in shear strength, and liquefaction in the alluvial environment. During the August 17, 1999, Kocaeli earthquake, much structural damage caused by liquefaction in saturated alluvial soils was reported. It was understood that liquefaction occurred in fine-grained soils and sands (Bol et al., 2010; Bray et al., 2004).

In evaluating liquefaction in fine-grained soils, the decision is made by evaluating the physical properties of the relevant soil or based on the laboratory test results. The disadvantages of laboratory experiments include limited accessibility, increasing costs, time-consuming procedures, and needing experienced operators to prevent them from being preferred in routine applications. In addition, difficulties in undisturbed

sampling, which are also seen in silty soils, similar to sands, create obstacles to providing initial stress conditions in the tests. There are many criteria in determining the liquefaction potential of fine-grained soils based on the physical properties of the relevant soil. In the liquefaction analysis of a fine-grained soil, physical properties of the investigated soil such as liquid limit, clay percentage, plasticity index, and w_n/w_L ratio representing in-situ soil consistency are taken into consideration.

One of the harmful effects of earthquakes on soils is liquefaction. Liquefaction is defined as the rapid increase in pore water pressures and reducing effective stresses to zero with a dynamic effect in saturated loose sandy soils (Kramer, 1996). The 1964 Niigata, Japan, and 1964 Great Alaska earthquakes focused scientists on liquefaction and played a significant role in developing liquefaction research. Seed and Idriss (1967), Lee and Seed (1967) observed two types of behavior depending on the relative density in undrained experiments performed under cyclic loads on clean sand samples in the laboratory. In loose sands, under cyclic loads, the pore water pressure suddenly increased and became equal to the effective stress, and as the soil liquefied, it lost its shear

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Table 1
Different criteria from the literature.

Reference	w_L (%)	Clay (%)	I_p (%)	w_n/w_L	D_{50} (mm)	I_L	σ'_{vo}
Wang (1979)	–	≤ 15 – 20	–	> 0.90	–	–	–
Jennings (1980)	–	$< 10^*$	< 10	–	> 0.02	–	–
Seed (1982)	< 35	$\leq 15^*$	–	≥ 0.90	–	–	–
Polito (1999)	≤ 30	–	≤ 10	–	–	–	–
Andrews & Martin (2000)	< 32	≤ 10	–	–	–	–	–
Polito & James (2001)	< 25	–	< 7	–	–	–	–
Seed et al. (2003)	< 37	–	< 12	> 0.80	–	–	–
Bray & Sancio (2006)	–	–	≤ 12	≥ 0.85	–	–	–
Bol et al. (2010)	< 33	< 10	–	–	> 0.02	> 0.90	–
Pathak & Purandare (2016)	–	≤ 15	≤ 15	≥ 0.90	> 0.02	–	150

strength by showing great deformation. On the other hand, in saturated dense sands, although the pore water pressure reaches a value equal to the effective stress at one stage of the loading cycle, with the expansion of the soil, the pore water pressure decreases, and the sample gains resistance against cyclic load. This phenomenon is called “pre-liquefaction.”

Although many studies have been conducted on the liquefaction of sands, studies on fine-grained soils’ liquefaction are relatively few. As a result of the liquefaction of silt soils in Adapazarı, especially in the 1999 Kocaeli earthquake, studies in this direction began to intensify in Turkey (Sancio et al., 2002; Bray et al., 2004; Bol et al., 2010).

SPT, CPT, and shear wave velocity (V_s) measurements can be used to determine the liquefaction potential of coarse sandy soils by using the cyclic stress (Andrus and Stokoe, 2000; Seed and Idriss, 1971; Youd and Idriss, 2001). In this method, a safety factor is obtained by comparing the shear stress created by the earthquake with the resistance of the soil. The result is obtained by simply dividing the resisting forces by the driving forces and correcting the earthquake magnitude.

Seed et al., 1985 stated that it would not be appropriate to use the cyclic shear stress method in determining the liquefaction potential of fine-grained soils where the cyclic shear stress method can be used for clean sands and silty sands. Liquefaction potential in fine-grained soils is generally determined by considering the physical properties of the soils.

Different researchers have suggested using some physical properties of soils to determine the liquefaction potential of fine-grained soils. It has been stated that the liquefaction potential of soils can be estimated by using parameters such as liquid limit (w_L), clay content (C%), water

content (w_n), liquidity index (I_L), and average grain size (D_{50}). Different liquefaction criteria use these parameters with varying limit values in the literature.

After the earthquakes in China, Wang (1979) and Seed et al. (1983) developed the methods, commonly known as the “Chinese Criterion”, and “Modified China Criterion” respectively, to decide if the liquefaction phenomena are possible. In these methods, for a fine-grained soil to liquefy, the liquid limit (w_L) should be less than 35, the clay content should be less than or equal to 15 %, and the water content (w_n) should be greater than $0.9 \times w_L$.

Koester (1994) stated that for liquefaction to occur, the liquid limit (w_L) must be less than 36, the clay ratio must be less than or equal to 10 %, and the water content must be more significant than $0.9 \times w_L$.

In their studies, Finn et al. (1994), for liquefaction to occur, stated that the liquid limit (w_L) should be less than 34 and the clay ratio should be less than or equal to 10 %.

Andrews and Martin (2000) stated that clay content and liquid limit could be considered “key” parameters are separating liquefiable and non-liquefiable soils. They said that if the liquid limit (w_L) is less than 32 and the clay ratio is less than or equal to 10 %, the soil is liquefiable, and if the soil meets any of these conditions, additional tests should be performed.

Bol et al. (2010) investigated the fine-grained soils in Adapazarı, where liquefaction or no liquefaction occurred. For liquefaction to occur in silt environments under groundwater level and at $M_w > 7$ conditions: (i) the liquid limit (w_L) is less than or equal to 33, (ii) the clay content is less than 10 %, (iii) liquidity index (I_L) or w_n/w_L ratio for NP soils should be greater than 0.9, (iv) the average grain size D_{50} should be greater than 0.02 mm. However, they revealed that soils with a value of $25 < w_L < 33$ and soils with a 10–15 % clay content and an average grain size of between 0.02 mm and 0.06 mm should be subjected to additional testing.

Some researchers, such as Polito (1999), Seed et al. (2003), and Bray and Sancio (2006), have developed criteria that consider the plasticity index of soils in the determination of liquefaction. Since liquefiable soils are generally “non-plastic (NP)” that do not show plastic properties or the plasticity index is difficult to determine, it is difficult to use these criteria in an extensive area in practice.

The criteria based on the physical properties proposed by the researchers are summarized in Table 1.

As it can be understood from the literature study, most researchers express a test region, and it is emphasized that additional tests should be done to determine this region. This is a reasonable suggestion, but it is a complicated and expensive method to bring undisturbed samples, especially silty soils, from the field. When sampling with the standard procedure, silty soils are heavily disturbed during insertion and removal from the tube.

It is not easy to determine the liquefaction potential of soils

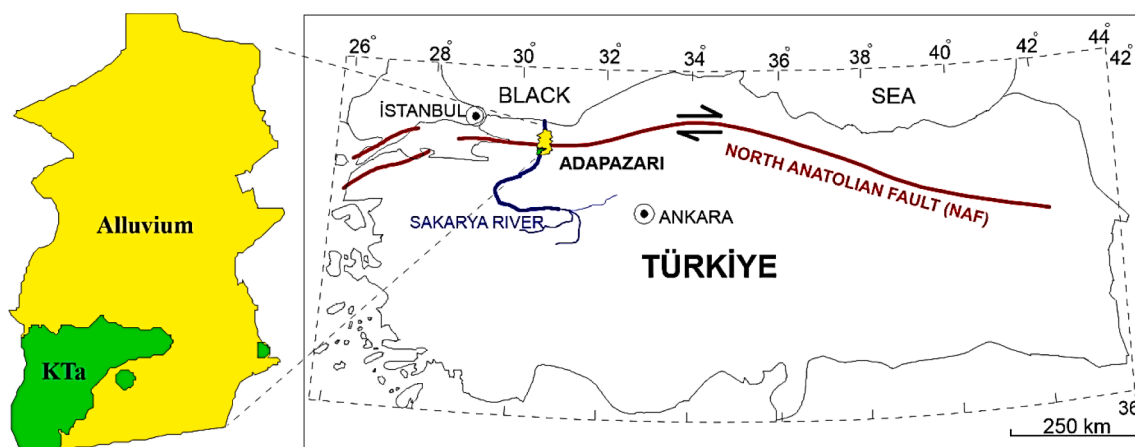


Fig. 1. The location of the study area.

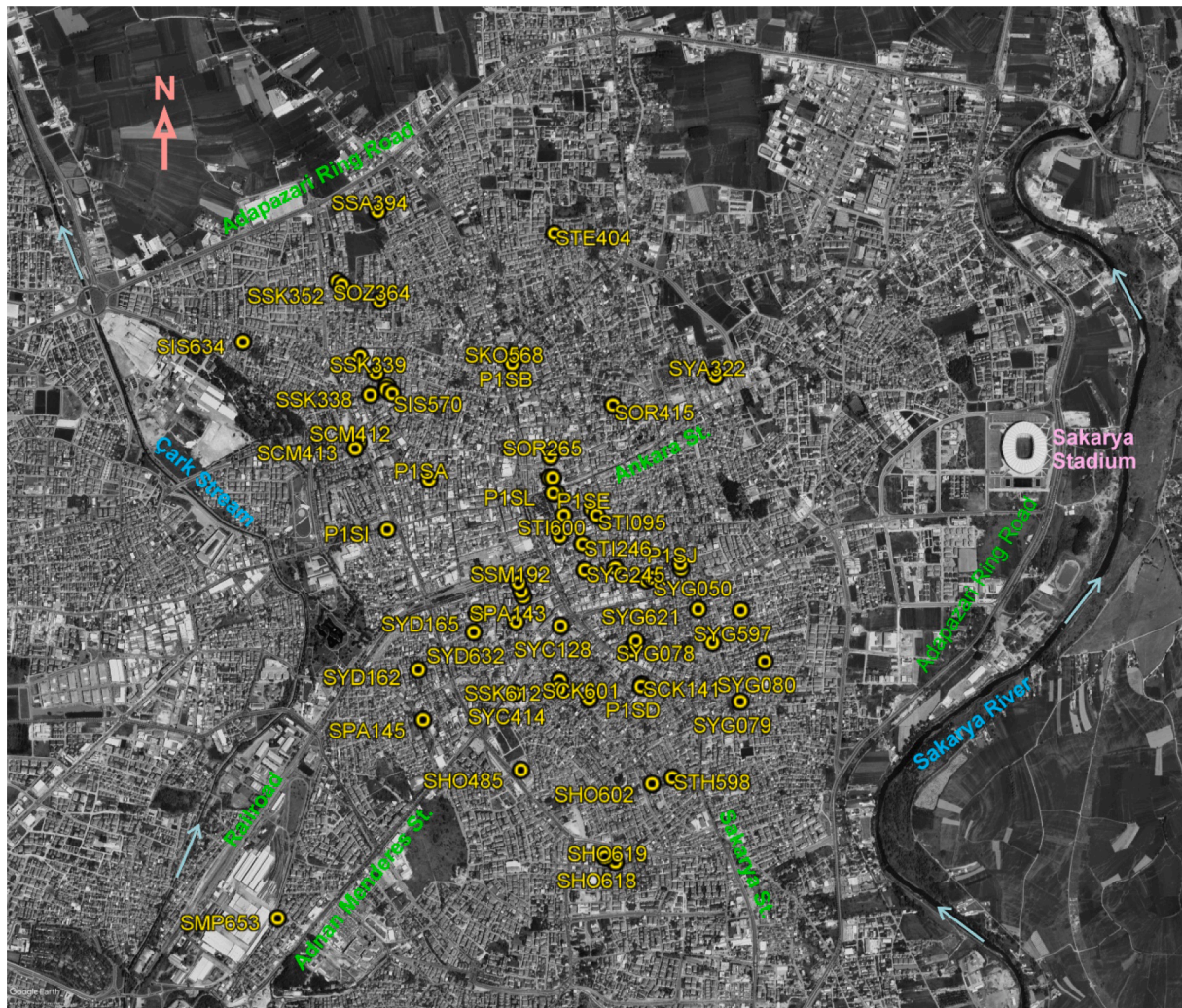


Fig. 2. Adapazari city centre and research points of the study.

Table 2

Attributes and descriptions (Soil properties).

Abbreviation	Description	Units	Data Type
Clay	Soil Particle Diameter is < 0.002 mm	percent (%)	Continuous
w_L	Liquid Limit	percent (%)	Discrete
IP	Plasticity Index	percent (%)	Discrete
w_n	Natural Water Content	percent (%)	Continuous
D_{50}	Mean Particle Size	mm	Continuous
FC	Fines Content	percent (%)	Continuous
Depth	Soil Depth	m	Continuous
Liq	Liquefaction	–	Binary

Table 3

Summary statistics of the dataset.

	Clay	w_L	IP	w_n	D_{50}	FC	Depth
mean	21.6	34.56	7.79	32.73	0.03	83.8	5.17
std	13.84	13.96	12.41	5.6	0.02	14.17	2.93
min	3	5	0	13	0	51	1
max	73	95	66	52	0.1	100	16.2

according to criteria based on physical properties due to the nonlinear relations. At this point, several “Machine Learning” (ML) methods are applied to determine soils’ liquefaction potential. Although it is given by comparing machine learning algorithms, the prominent algorithms and

studies can be given as follows: Kumar et al. (2021) presented a deep learning (DL) model for reliable soil classification in liquefaction determination. The applicability of the DL model was tested with emotional backpropagation neural networks (EMBP). Tung et al. (1993) present a computer model for estimating soil liquefaction potential using developed neural networks. The model is built with data sets obtained from past events, and it is stated that it can be used to understand future events. García et al. (2012) presented a machine learning scheme to evaluate the liquefaction potential of soils based on geotechnical, geometric, and seismic load parameters. A relatively large database of CPT and vs measurements and field liquefaction performance observations of historical earthquakes are analyzed. This database creates a nonlinear environment where liquefaction can be predicted using neural networks and classification trees. Ahmad et al. (2021) investigated the performance of four machine learning (ML) algorithms by cone penetration test (CPT) based on field case history records to evaluate the earthquake-induced liquefaction potential of the soil. Hu (2021) created two Bayesian network models to predict soil liquefaction based on dynamic penetration testing and shear wave velocity databases. The models made were compared with the existing models, and it was stated that they gave a good performance. Some other studies can be found in the literature, such as artificial neural networks (Park et al., 2020; Tung et al., 1993); deep-learning (Feng et al., 2019); decision tree (Ahmad et al., 2021; García et al., 2012), support vector machines (Goh & Goh, 2007; Samui & Sitharam, 2011), multiple linear regression (Makasis et al., 2018), super learner (Taghizadeh-Mehrjardi et al., 2021),

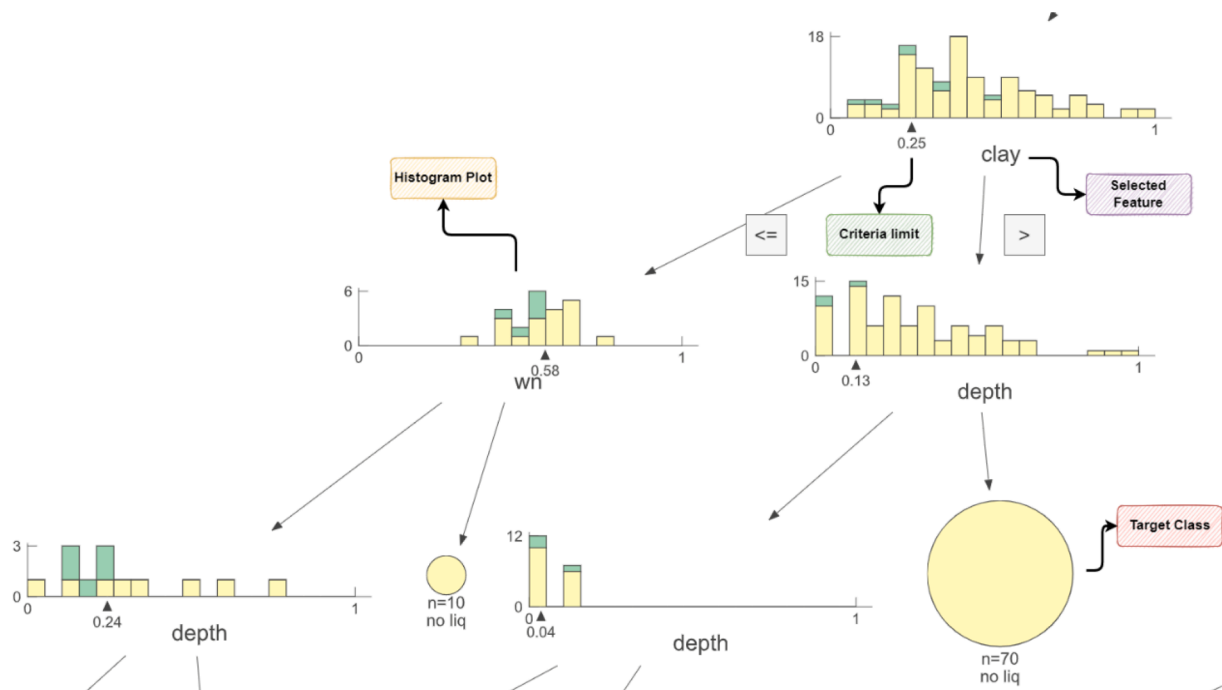


Fig. 3. A sample representation of the decision tree.

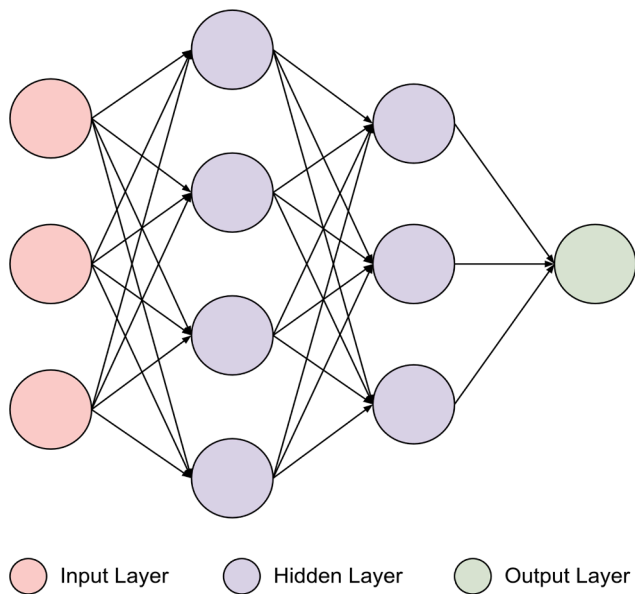


Fig. 4. Network Architecture of an MLP.

ensemble learning (Alobaidi et al., 2019).

This study aims to understand the gray region between the different transition zones proposed liquefiable-non-liquefiable zone. ML models can be used to discover linear or nonlinear relationships. In this study, the advantages and effectiveness of ML models, including decision trees (DT), logistic regression (LOGREG), support vector machines (SVM), k-nearest neighbors (KNN), stochastic gradient descent (SGD), random forest (RF), and artificial neural network (ANN) for the prediction of soil liquefaction potential were investigated. Algorithms were coded in Python and applied to the dataset taken from the Adapazari/Turkey region after the 1999 Kocaeli earthquake. The main contribution of our work includes (i) Finding the best predictive model for soil liquefaction. (ii) An original data set (never used for machine learning) in Adapazari data

was modeled by ML models. (iii) Different ML models were tried for different parameters, and the best model was the decision tree algorithm. (iv) The model can be used in Adapazari soil samples. This paper is organized as follows: Section 2 presents the methodology and the proposed approaches, including data collection and characteristics. Also, it explains the details of the different machine learning methods. Section 3 presents the experimental results and performance comparison of the ML models. Finally, Section 4 presents the conclusions and possible future works.

2. Materials and methods

2.1. Study area and dataset

The field data results of many drillings and seismic cone penetration tests (SCPT) were collected in the dataset covering the entire city of Adapazari, Turkey. In addition, classification and strength properties tests were carried out on the samples from the drillings and transferred to the database (Onalp et al., 2001; Bol et al., 2008, 2010). Bray et al. (2004) presented critical layers in Adapazari where soil damage due to liquefaction was observed after the 1999 Kocaeli earthquake. These layers' soil properties were taken from the report published on Pacific Earthquake Engineering Research Center (PEER) website and added to the database (Bray et al., 2001).

Soils can be formed in many different environments, including the main titles of residual and transported soils. Soil types that cause problems in geotechnical are generally soils formed by transport with thick layers. This type of soils can also be formed in very different facies and sub-facies environments. For example, moraines carried by glaciers, dunes formed by winds, talus formed by gravity and finally alluviums carried by water. Alluviums carried by water are considered to be of fluvial origin. In particular, the meandering river-based fluvial environments from which the soils used within the scope of this study originate carry very different sub-facies. For example, abandoned riverbed, set top deposits, cravasse splay deposits, oxbow lake deposits and flood plain deposits. The soils used in this study are also fluvial-origin flood plain sediments. The Adapazari soils, from which the samples used in this study were taken, are flood plain sediments of fluvial

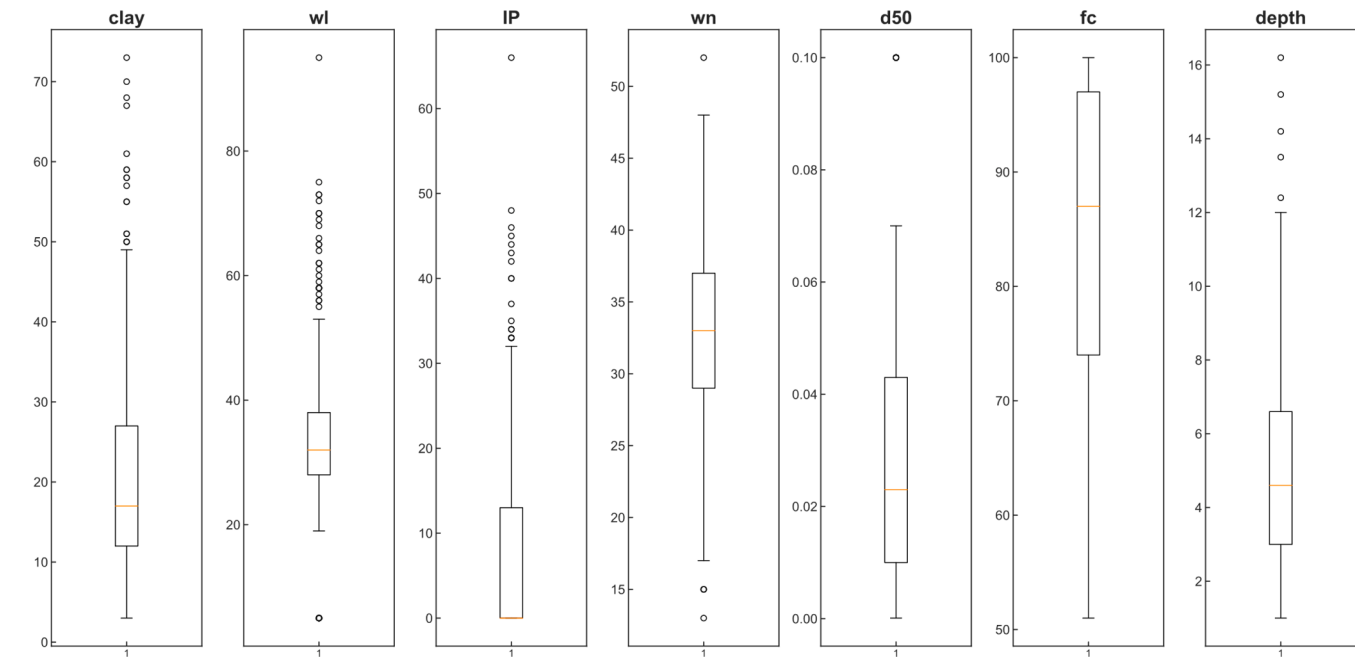


Fig. 5. Boxplots indicating the distribution of the features.

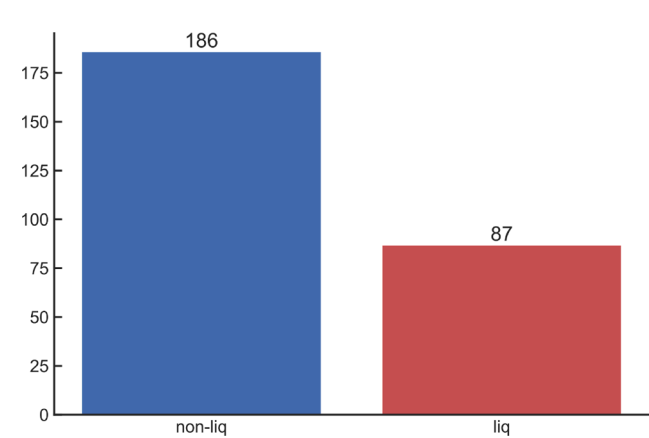


Fig. 6. Target value frequencies.

origin.
A large part of Adapazarı and its surroundings are formed by quaternary alluvial deposits containing gravelly and silty sands brought by

		Actual Values	
		1: liq	0: no liq
Predicted Values	1	TP	FP
	0	FN	TN

Fig. 8. Confusion matrix of the soil liquefaction potentials.

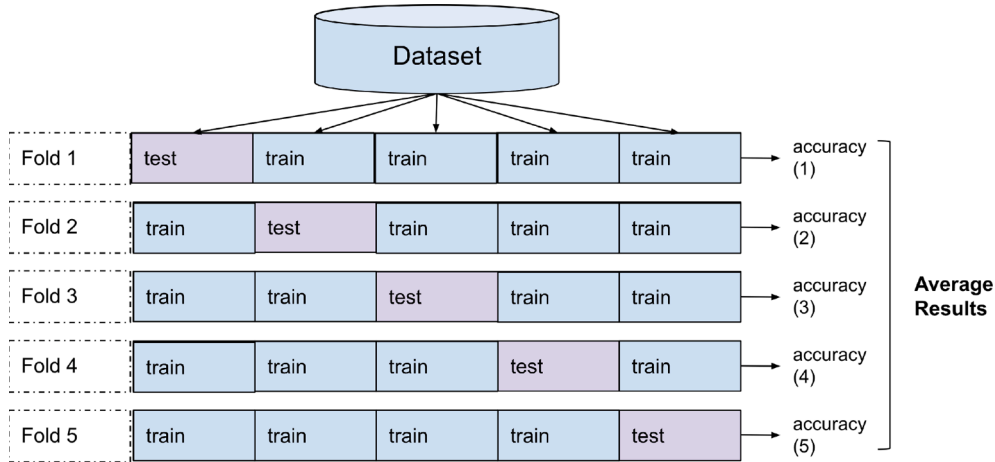


Fig. 7. 5-fold cross-validation method.

Table 4

Correlation matrix of the soil features.

	Clay	w _L	IP	w _n	D ₅₀	FC	Depth
w _L	0.691						
IP	0.761	0.859					
w _n	0.409	0.481	0.464				
d50	−0.616	−0.572	−0.550	−0.367			
FC	0.536	0.620	0.534	0.481	−0.679		
Depth	0.139	0.131	0.132	−0.050	−0.109	0.104	
Liquefaction	−0.437	−0.413	−0.355	−0.148	0.344	−0.370	−0.245

Table 5

Selected hyperparameters of each algorithm.

Model	Parameters	No. of Exp.
DT	max_depth: [3,4,5,...,20], min_samples_leaf: [3,4,5,...,20], criterion: [gini, entropy], splitter: [best, random], max_features: [auto, sqrt, log2]	17,340
LOGREG	penalty: [l1, l2, none], C: np.linspace(-100, 100, 50)	4500
SVM	fit_intercept: [True, False], solver: [newton-cg, lbfgs, liblinear] kernel: [rbf, linear], C: np.linspace(-100, 100, 50) gamma: np.linspace(0, 100, 20)	10,000
KNN	n_neighbors: [1,5,10,...,100], weights: [uniform, distance], leaf_size: [20, 25, ..., 100], p: [1, 2], algorithm: [auto, ball_tree, kd_tree, brute]	25,600
SGD	loss: [hinge, log, modified_huber, squared_hinge, perceptron], penalty: [l1, l2, elasticnet], alpha: np.linspace(0.001, 100, 50) learning_rate: [constant, optimal, invscaling, adaptive], eta0: [1, 10, 100]	45,000
RF	criterion: [gini, entropy], max_depth: [2,3,4,...,10], min_samples_split: [2,3,4,...,10], min_samples_leaf: [2,3,4,...,10], n_estimators: [2,3,4,...,10], max_features: [auto, sqrt, log2]	122,880
ANN	hidden_layer_sizes: [(2,),(3,),(4,),(2,2),(2,3),(2,4),(3,3),(3,4),(4,3),(4,4),(5,5),(6,6)], activation: [tanh, relu, logistic], solver: [sgd, adam], alpha: [0.0001, 0.001, 0.01, 0.1, 0.05], learning_rate: [constant, adaptive]	3300

the Sakarya river. Different facies are observed to varying depths in different parts of the Adapazari plain. A small section in the city's southwest was founded on the cretaceous Akveren formation. The formation consists of claystone, sandstone, limestone, and marl layers. The densely populated areas are on the alluvial layers (Fig. 1). Adapazari city centre and research points of the study are shown in Fig. 2.

In the initial dataset, presented in Table 2, includes soil samples taken from different points of the Adapazari region and 8 soil features. A column on whether the soil properties used caused soil liquefaction is also included in the data set with the binary values [0,1]. The value of "0" denotes non-liquefied, and "1" denotes liquefied soils, which is known as the "target" variable in data science applications. The binary criterion of liquefaction/no-liquefaction was judged primarily based on surface manifestation of liquefaction, such as sand boils, ground settlement, and lateral spreading (or lack thereof), and in some cases (such as those reported by Bray et al. 2004; Bol et al. 2010), critical layers were identified by field observations supplemented with detailed dynamic finite element analyses or confirmed by multiple existing liquefaction

evaluation methods.

In the data preprocessing stage, the data missing any feature was detected and removed from the data set. At the same time, inconsistent data in the data set (- values, non-numerical, different typed data with the same meaning) was cleaned (cleansing). After the pre-processing stages, there is very little data left in the dataset that is inconsistent or noisy. Having missing values in a data set reduces the model performance (Kang, 2013). The statistical summary of the feature sets includes the mean, standard deviation (std), minimum (min), and maximum (max), which are presented in Table 3.

2.2. Machine learning methods

Machine learning (ML) models, a subset of artificial intelligence, can be divided into three subgroups. These are supervised learning, unsupervised learning, and reinforced learning. Supervised learning is used when the target variables in the data set are known. In unsupervised learning, the values of the target variables are initially unknown (El Naqa & Murphy, 2015). The target variable discussed in this study, i.e., the potential for soil liquefaction will be given to the model. Therefore, supervised learning algorithms will need to be used. Supervised learning algorithms can be collected under regression and classification studies. The target variable in this study is binary, so the study will be treated as a classification study. In most studies, it can be determined by trialing different algorithms that will work better. The quality and quantity of data can be determined by looking at the relationship of characteristics in the data with each other to select appropriate machine learning algorithms (Zhang, 2020). After the relationships in the data are revealed, the data set is trained with parametric methods such as logistic regression, artificial neural networks, naive Bayes, or non-parametric methods such as k-nearest neighbors algorithm, decision trees, and support vector machines (Bonaccorso, 2017). If linear models can be used among the features in the data set, the model will work better with linear machine learning algorithms. The approaches to this study include both parametric and non-parametric methods, which are outlined below.

2.2.1. Decision trees

Decision tree algorithm (DT) is used for classification tasks in machine learning, and it can work with both categorical and continuous data. Thus, it has a more widespread use (Charbuty & Abdulazeez, 2021). DTs enable modeling in a tree structure that extends from the root to the leaves. The creation of the tree begins from the root. When determining the feature on the root node, the Gini index or entropy values are examined. The impurity status of the feature denotes a degree of inhomogeneity values. If the probability of a condition is p_i , the entropy value (E) is calculated as follows (Breiman et al., 1984; Kingsford & Salzberg, 2008; Quinlan, 2014; Rokach & Maimon, 2005):

$$E = - \sum_{i=1}^m p_i \log(p_i) \quad (1)$$

The formula for the Gini index (G) is estimated as:

$$G = 1 - \sum_{i=1}^m p_i^2 \quad (2)$$

These measures determine the impurity value of a feature. The feature selection process is completed by choosing the highest impurity

Table 6
Best hyperparameter combinations for DT algorithm.

criterion	max_depth	max_features	min_samples_leaf	splitter	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
gini	12	auto	3	best	0.857	0.821	0.679	0.714	0.821	0.779	0.069
	7	auto	5	random	0.786	0.714	0.750	0.786	0.821	0.771	0.036
	7	auto	3	best	0.821	0.786	0.714	0.714	0.786	0.764	0.043
	8	auto	3	best	0.857	0.821	0.679	0.750	0.714	0.764	0.066
	11	auto	3	best	0.857	0.714	0.679	0.714	0.821	0.757	0.069

Table 7
Best hyperparameter combinations for LOGREG algorithm.

C	fit_intercept	penalty	solver	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
10.204	True	l1	liblinear	0.857	0.786	0.750	0.714	0.750	0.771	0.048
2.041	True	l1	liblinear	0.821	0.786	0.750	0.714	0.750	0.764	0.036
2.041	False	l1	liblinear	0.821	0.786	0.786	0.750	0.679	0.764	0.048
10.204	True	l2	liblinear	0.821	0.786	0.714	0.750	0.714	0.757	0.042
10.204	False	l2	newton-eg	0.786	0.750	0.786	0.786	0.679	0.757	0.042

Table 8

Best hyperparameter combinations for SVM algorithm.

C	gamma	kernel	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
26.531	0.100	linear	0.857	0.786	0.750	0.714	0.750	0.771	0.048
55.102	0.100	rbf	0.821	0.786	0.714	0.786	0.714	0.764	0.043
10.204	0.100	rbf	0.750	0.714	0.786	0.821	0.750	0.764	0.036
91.837	0.100	rbf	0.857	0.786	0.714	0.714	0.714	0.757	0.057
14.286	0.100	rbf	0.750	0.714	0.786	0.786	0.750	0.757	0.027

values because the feature with a high impurity value provides the most information gain. After deciding which feature to branch out in the decision tree, the criteria are determined. Criteria are branched out to see which number is more significant than smaller. If there is a 100 % class under the branch, the representation in that branch is completed.

Fig. 2 is an example decision tree generated from the data in this study. Accordingly, clay has been selected as first branching feature. If clay is lower than 0.25, it should go to the left side (the w_n side). Otherwise, in the right side, if depth is lower than 0.13, it should go to the left side. Otherwise, as a result of the branching to the right, the target class is determined as non-liquefied based on 70 samples.

2.2.2. Random forest

Learning is carried out by using a single tree in DT. Ensemble methods work with more than one tree. DTs may involve overfitting. The random forest algorithm performs training by working with some decision trees to prevent overfitting (Breiman, 2001). The random forest algorithm chooses the best by rating decision trees. The random forest algorithm is an algorithm that can be used instead of decision trees to avoid overfitting in high-size data sets. Other models developed with decision trees are boosting algorithms. In this study, ensemble decision trees were applied with random forest algorithms.

2.2.3. Support vector machines

Support vector machines (SVMs) are an algorithm classified by drawing boundary lines between classes (Joachims, 1998) used for both regression and classification problems (Mavroforakis & Theodoridis, 2006). SVM ensures the maximum distance between classes where boundary lines are drawn. Boundary lines of support vector machines are sometimes linear and sometimes work with high-sized kernels to predict more features.

2.2.4. K-nearest neighbors

K-Nearest Neighbors algorithm (K-NN) is another non-parametric classification algorithm. It can be used in both classification and regression (Altman, 1992). Because the K-NN algorithm is not parametric, it provides an excellent opportunity for data that does not fit a model. The most crucial parameter in the K-NN algorithm is the number of k , which specifies how many neighbors a point belongs to a class. In this study, the k coefficient in the KNN algorithm was tried with different parameters.

2.2.5. Artificial neural network (multilayer perceptron)

Artificial neural networks or multilayer perceptron (MLPs) consist of 3 layers: input, hidden, and output layers. MLPs may not have hidden layers. This case means that a simple artificial neural network (ANN) has been used. Besides, hidden layers can be used to detect more complex relationships. The number of units in the layers is one of the most critical parameters of these algorithms. The layers are entirely connected. Fig. 3 shows a representative artificial neural network.

2.2.6. Logistic regression

Linear regression models are used if the output variables are continuous. Logistic regression models have been developed to use regression models in classification problems. Intercept and slope values are estimated as linear regression in logistic regression (Hosmer et al.,

2013). A function is needed for the predicted value to be intermittent, a logistical function. Thanks to the logistics function, output values in linear regression are discontinuous. The resulting categorical variables indicate a class. The formula of the logistics function is given in Eq. (3).

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}} \quad (3)$$

Where,

L : max value of the curve

k : logistic growth rate of the curve

x_0 : sigmoid's midpoint x value

2.2.7. Stochastic gradient descent (SGD)

Stochastic gradient descent (SGD) is an iterative procedure applied in many machine learning studies. The algorithm SGD mimics Polyak's heavy-ball method for the optimization problems (Polyak, 1964). SGD is a good and popular algorithm that has been implemented in well-known machine learning packages Pytorch (Ketkar & Moolayil, 2021) and Tensorflow (Dillon et al., 2017).

2.3. Data pre-processing

It is necessary to determine the practical factors in soil liquefaction to make the data set appropriate for machine learning applications. For this purpose, the factors determined by expert opinion are included in the data set. One essential pre-processing phase that improves the performance of machine learning algorithms is the scaling or normalization process. In this study, each feature is scaled with min-max scaling. Using Eqs. 4 and 5, the min-max scaling method scales the features to the range [0,1].

$$X_{std} = \frac{X - X_{min}(axis=0)}{X_{max}(axis=0) - X_{min}(axis=0)} \quad (4)$$

$$X_{scaled} = X_{std} \times (max - min) \quad (5)$$

here X is the original value, X_{min} and X_{max} the minimum and maximum value of the feature, respectively. X_{scaled} is the scaled value.

Boxplot graphs are drawn to detect outlier data (Fig. 4). Since the outlier data is not at the extremes, it was decided not to remove it from the model with expert opinion.

There are 186 data points for non-liquefied and 87 data points for liquefied soils in the data set, shown in Fig. 5. If dataset of machine learning applications is imbalanced (the number of observations in one target class is significantly lower than the number of observations in another class), this should be considered in training and test sets. F1-score is the most appropriate metric for calculating performance of the model (Eq. 9). If there are excessive classes in the training set, poor performance can be observed in the test set. Distributed data sets can be determined close to each other in training and test sets to eliminate this situation. In this study, since the entire data set is used in both training and verification using cross-validation, there is no need for a distributed data set preference close to each other.

The best model can be developed with cross-validation without relying on the training and test set in the data set. If only one training and test set is used, a good performance obtained by adhering to the

Table 9
Best hyperparameter combinations for KNN algorithm.

algorithm	leaf_size	n_neighbors	p	weights	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
auto	20	11	1	uniform	0.821	0.786	0.750	0.750	0.750	0.771	0.029
auto	20	6	1	distance	0.857	0.714	0.714	0.750	0.821	0.771	0.058
auto	65	11	1	uniform	0.821	0.786	0.750	0.750	0.714	0.764	0.036
auto	20	21	1	distance	0.750	0.786	0.714	0.821	0.750	0.764	0.036
auto	20	11	1	distance	0.786	0.786	0.714	0.750	0.750	0.757	0.027

Table 10
Best hyperparameter combinations for SGD algorithm.

alpha	eta0	learning_rate	loss	penalty	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
0.001	10	invscaling	log	elasticnet	0.786	0.786	0.786	0.821	0.786	0.793	0.014
0.001	10	invscaling	hinge	l1	0.821	0.786	0.786	0.786	0.786	0.793	0.014
0.001	10	invscaling	log	l2	0.786	0.786	0.786	0.786	0.786	0.786	0.000
0.001	1	invscaling	modified_huber	l2	0.821	0.786	0.750	0.821	0.714	0.779	0.042
0.001	100	adaptive	modified_huber	l1	0.821	0.786	0.786	0.750	0.750	0.779	0.027

Table 11
Best hyperparameter combinations for ANN algorithm.

activation	alpha	hidden_layer_sizes	learning_rate	solver	split0_test_score	split1_test_score	split2_test_score	split3_test_score	split4_test_score	mean_test_score	std_test_score
tanh	0.0001	(5, 5)	constant	adam	0.750	0.750	0.786	0.786	0.750	0.764	0.017
tanh	0.1	(5, 5)	constant	adam	0.750	0.750	0.786	0.750	0.750	0.757	0.014
relu	0.001	(5, 5)	constant	adam	0.750	0.750	0.750	0.714	0.750	0.743	0.014
relu	0.0001	(5, 5)	constant	adam	0.714	0.750	0.750	0.714	0.750	0.736	0.017
relu	0.0001	(3, 4)	constant	adam	0.750	0.750	0.750	0.714	0.679	0.721	0.027

training set may not be generalized to the entire test set or data set. If the model produces results that are very close to the data in the training set but produce far-from-the-right results in the test set, the model may have 'overfitting' memorization. If the model has low performance in the training set and high performance in the test set, which means the model is simple, called 'underfitting'. The generalization of machine learning models is possible with hyperparameter optimization and cross-validation methods. In cross-validation, the data set is divided into k subsets, then the $k - 1$ set is used for training, and the other set is used for testing. In this study, the cross-validation data set was divided into five parts. In other words, 20 % validation and 80 % of the data set are used for training with each validation. Fig. 6 illustrates how to separate the data set into a training and test set in the cross-validation phase.

2.3.1. Performance metrics

The model with the best average performance value in cross-validation was used for modeling the data set. In the validation of the model, the data set split %80, %20 for the training and testing process, respectively. Moreover, the training accuracy, test accuracy, and accuracy of the entire data set were saved. Performance metrics were used to evaluate the performance of the model. Since the problem addressed is a binary classification problem, the confusion matrix consisting of a 4-cell window can be used for performance comparisons. Fig. 7 shows the confusion matrix for this study.

TP (True Positive): Correct predictions of liquefied soil.

TN (True Negative): Correct predictions of non-liquefied soil.

FP (False Positive): Incorrect predictions of liquefied soil.

FN (False Negative): Incorrect predictions of non-liquefied soil.

Accuracy is the most common performance metric used for binary classification problems. Another performance metric is the true positive rate (TPR) or recall, true positive predictions. Accuracy and the TPR are calculated using Eqs. 6–7:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

True Negative Rate (TNR) or Precision is the proportion of true negative predictions, which can be calculated by Eq. 8:

$$Precision = \frac{TN}{TN + FP} \quad (8)$$

F1-score or F-score or F-measure is the precision and recall weighted average. F1-score can be calculated by Eq. 9:

$$F_1 = 2 \times \frac{precision \times recall}{precision + recall} \quad (9)$$

Area Under Curve (AUC), the Receiver Operating Characteristic (ROC) represents true positive and false-positive rates for different threshold values.

2.3.2. Feature selection

In the data preparation, incorrect data detection and cleaning, outlier data analysis, and incomplete data transactions were carried out. The data deemed contrary was removed from the data set by taking expert opinion to clean up the outlier data. In addition, the missing data has been removed from the data set because it is predicted that model performance will decrease due to filling in the missing data with one of the missing data filling methods.

Scatter plots are given in Fig. 8 to show the distributions between properties and the target variable. As shown in Fig. 8, some features do not exhibit parametric distribution. When looking at the different feature values for liquefied and non-liquefied soils together, it was observed that it is impossible to make a linear distinction. Such situations will affect the performance of machine learning algorithms. It has

Table 12
Performance comparisons of the models.

Models	Accuracy			precision		recall		f1-score	
	Train	Test	All	0 (non-liq)	1 (liq)	0	1	0	1
DT	0.90	0.89	0.96	0.90	0.89	0.96	0.77	0.93	0.83
LOGREG	0.83	0.71	0.88	0.83	0.71	0.88	0.62	0.86	0.66
SVM	0.83	0.71	0.88	0.83	0.71	0.88	0.62	0.86	0.66
KNN	0.86	0.67	0.84	0.86	0.67	0.84	0.71	0.85	0.69
RF	0.91	0.77	0.89	0.91	0.77	0.89	0.80	0.90	0.79
ANN	0.79	0.72	0.91	0.79	0.72	0.91	0.47	0.85	0.57
SGD	0.78	0.79	0.95	0.78	0.79	0.95	0.43	0.85	0.55

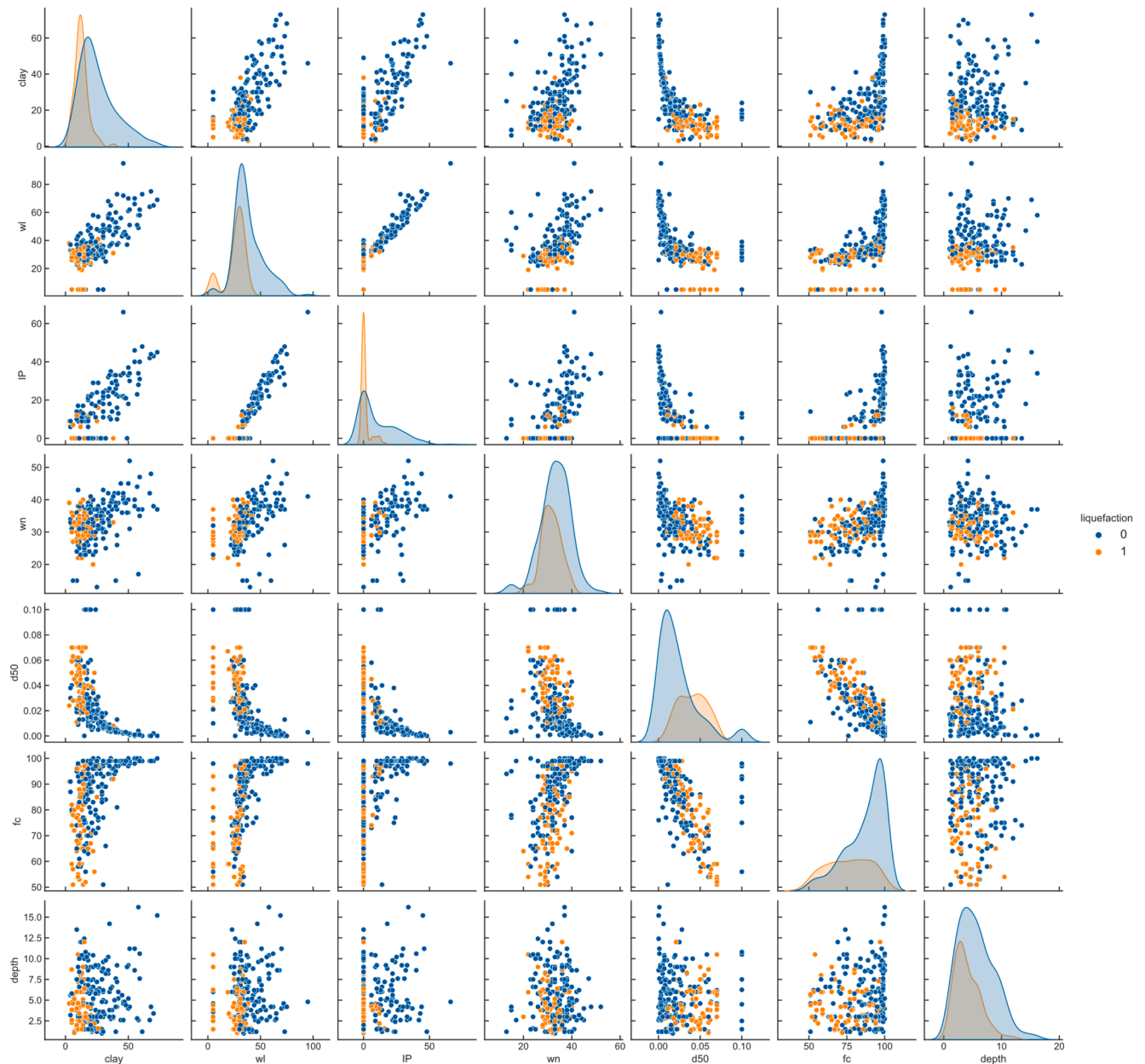


Fig. 9. Relation between features.

been observed that the values of 0, especially in the IP feature, are in high numbers. In the laboratory, the plastic limit is difficult to measure on silty soils and is very dependent on the operator. In addition, the repeatability of the plastic limit test is one of the most difficult among geotechnical laboratory tests. Therefore, the IP feature is not considered

in many liquefaction criteria.

The correlation matrix was used for interactions between features (Table 4). If the correlation between the features is high, one of the features must be removed from the data set. In this study, it was aimed to work with as many features as possible. Therefore, feature with the

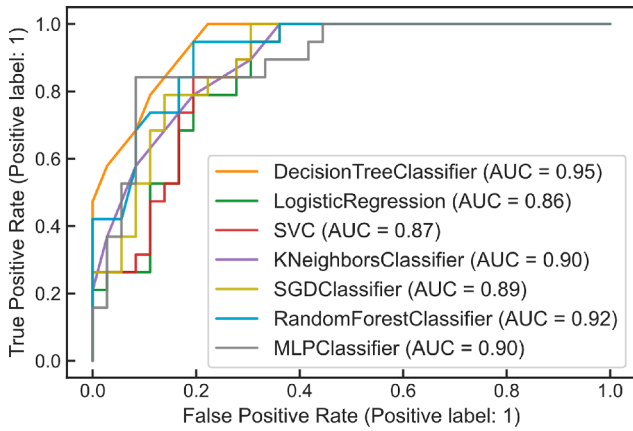


Fig. 10. ROC Curve for both test and train sets of the models.

highest correlation(IP feature with a correlation of 0.86) is removed from dataset. So, the threshold value for this study was set as 0.85.

3. Results and discussion

The Scikit-Learn library was used in the Python programming language for machine learning models and codes developed for performance calculations in this study (Pedregosa et al., 2011). The experiments were done on a laptop with Intel(R) Core(TM) i7-5600U CPU @ 2.60 GHz and 12 GB RAM running 64-bit Windows. The data set discussed in the study is trained with 5-folds cross-validation. The average of the cross-validation results was taken and sorted according to different parameters using the Grid Search technique as an optimization method. Table 5 provides value ranges for the parameters used and the total number of experiments of each algorithm.

Models have been developed with the best parameters obtained due to cross-validation. Five of the best combinations obtained by Decision Tree (DT), Logistic Regression (LOGREG), Support Vector Machines (SVM), k-Nearest Neighbor (KNN), Stochastic Gradient Descent (SGD), Random Forest (RF), and Artificial Neural Network (ANN) cross-validation are given as in Tables 6-11. The best parameter combination obtained with the parameters specified in the table will then be tried on the entire data set and added to the study's outputs. Some hyperparameters combinations produced the same results. One of these combinations is presented in summary. The information presented in the tables shows an example of the best results. Performance evaluations

were made according to the average test accuracy score. CPU times of the algorithms were given as 64.5 s 44.2 s 58.4 s 122.9 s 135.2 s 320 s respectively.

Models have been developed with the best parameters obtained due to cross-validation. The results obtained when the model has tried again in the randomly selected 20 % test set with the algorithm that gives the best result due to cross-validation are summarized in Table 12. The best overall accuracy for all dataset is 90 % was reached in the DT classifier. After the DT classifier, the RF classifier, a decision tree-based algorithm, comes in second place with an 86 % accuracy rate. According to table, DT obtained best accuracy rate in also precision (91 %), recall (96 %), and f1-score(93 %, 83 %). In addition, the results in which the RF algorithm can also be highlighted. RF obtained the best results on precision and recall rate.

Another performance metric that is noted in this study is the ROC (Receiver Operating Characteristic) in which true positive rate is usually plotted on the Y axis, whereas false positive rate is plotted on the X axis. This suggests that the “ideal” point is in the upper left corner of the figure, with a false positive rate of zero and a true positive rate of one (scikit-learn, 2022). The true positive rate represents the percentage of liquefied soils that the model predicted as liquefied. Vice versa, false positive rate is the percentage of non-liquefied soils, but the model predicted them as liquefied soil. This metric summarizes the goodness of fit of a classifier. ROC curves for both training and test sets are given in Fig. 9. The best feasible prediction approach would result in a point in the upper left corner of the ROC space. A random guess would result in a

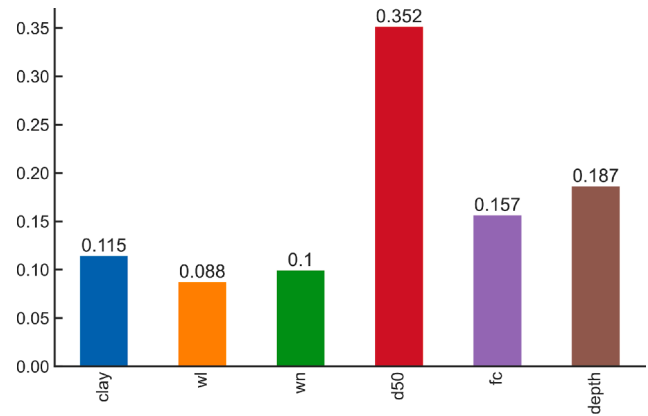


Fig. 12. Feature importance generated from DT model.

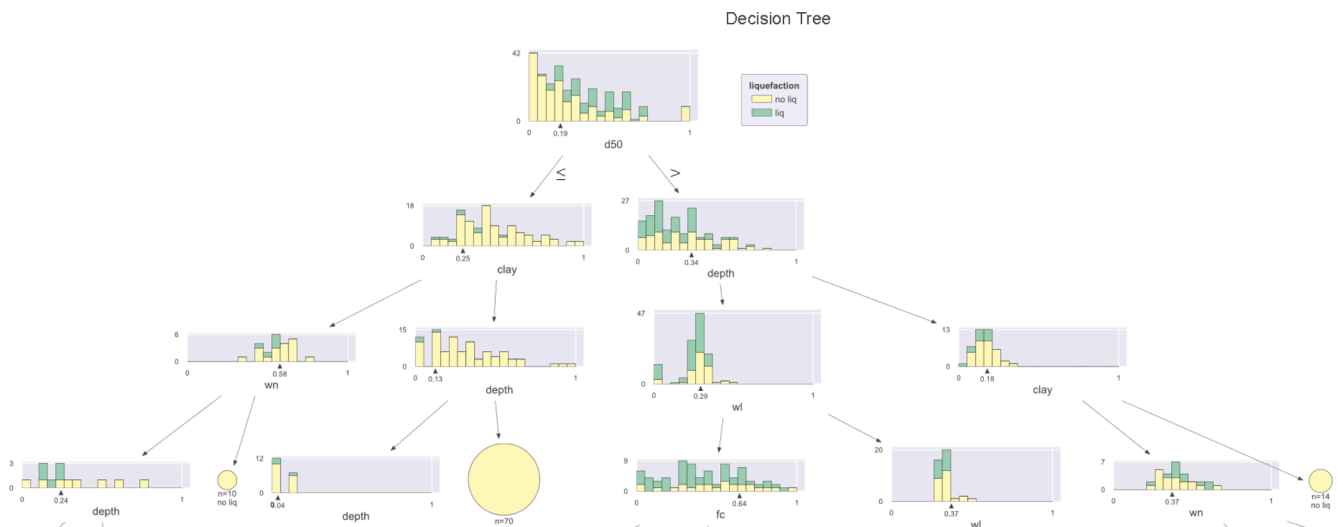


Fig. 11. Best decision tree model visualization.

point running diagonally from the bottom left to the top right corners. Looking at the ROC curves, it can be seen that the DT classifier again gives the best results with 97 % in the training set and 95 % in the test set best on various parameters. The RF classifier comes in second place.

According to the results given, it is concluded that the DT classifier can be used in the estimation of the potential for soil liquefaction. In the classification made with the decision tree, 178 non-liquefaction soil cases were correctly estimated. Conversely, 8 cases were incorrectly estimated. Sixty-seven samples from non-liquefaction soils were accurately estimated. By contrast, 20 cases were incorrectly estimated. The DT classifier is an algorithm that is easy to read and interpret. One of the outputs of the algorithm is the rules of decision. Decision rules can be created to predict new data. A few of the decisions considered important are shown in Fig. 10.

Moreover, D_{50} feature is found as the first splitting attribute, as shown in Fig. 11. That's why D_{50} indicates the most important feature in the dataset. As mentioned, D_{50} had the strongest effect on the liquefaction prediction in the case of DT model, while the depth feature had the second strongest effect. The relative importance of each feature for predicting soil liquefaction is given in Fig. 12.

The limit values were calculated by inverse scaling so that the results could be evaluated with their actual values. The important decision rules are summarized as follows:

- if ($d_{50} \leq 0.01948$) and ($clay > 20.5$) and ($depth > 2.9$) then class: no liq (proba: 100.0 %) | based on 70 samples
- if ($d_{50} > 0.01948$) and ($depth \leq 6.1528$) and ($w_l \leq 31.45$) and ($fc \leq 82.507$) and ($depth \leq 4.4504$) and ($d_{50} > 0.03296$) and ($clay \leq 14.48$) and ($d_{50} \leq 0.0595$) then class: liq (proba: 100.0 %) | based on 17 samples
- if ($d_{50} > 0.01948$) and ($depth > 6.1528$) and ($clay > 15.53$) then class: no liq (proba: 100.0 %) | based on 14 samples
- if ($d_{50} \leq 0.01948$) and ($clay \leq 20.5$) and ($w_n > 0.577$) then class: no liq (proba: 100.0 %) | based on 10 samples
- if ($d_{50} \leq 0.01948$) and ($clay > 20.5$) and ($depth \leq 1.5472$) then class: no liq (proba: 100.0 %) | based on 9 samples
- if ($d_{50} > 0.01948$) and ($depth \leq 6.1528$) and ($w_l > 31.45$) and ($w_l \leq 38.48$) and ($w_n \leq 38.506$) and ($d_{50} \leq 0.055$) and ($fc \leq 93.483$) and ($w_l \leq 35.95$) and ($fc > 88.485$) then class: no liq (proba: 100.0 %) | based on 6 samples
- if ($d_{50} \leq 0.01948$) and ($clay > 20.5$) and ($depth \leq 2.9$) and ($depth > 1.5472$) and ($d_{50} \leq 0.0059$) then class: no liq (proba: 100.0 %) | based on 6 samples

4. Conclusions

This study presents a model for soil liquefaction potential using seven well-known machine learning approaches: DT, LOGREG, SVM, KNN, RF, ANN, and SGD. The Adapazari data set is trained after being evaluated with expert opinion and passed through the pre-processing stages. In the training process, the results of the algorithms with different parameters were recorded. Thus, algorithms with the best parameters are provided with better results. When the algorithms used were sorted according to the quality of training of the entire data set, it was observed that the decision trees obtained the highest results with an accuracy 90 % success rate. The results obtained by the RF classifier were the second-best algorithm. The specified DT model resulted in 91 % accuracy in the randomly selected training set and 84 % accuracy in the test set. F1 score of the DT model reached 96 %. The rules, which previously stated whether there was liquefaction with expert opinion, were developed by machine learning methods. DT model in this study presents a good alternative for the prediction of liquefaction and gives more accurate results than the previous works. However, many methods in the literature offer a test site where the decision cannot be made as soon as it liquefies. Decision rules produced with decision trees are proposed in the results and discussion section. To sum up, soil

liquefaction decisions can be made using these rules. In addition, the importance of the features in the data set discussed was determined by decision trees. Accordingly, the feature ranking was discovered as D_{50} , depth, FC, clay, w_n , w_l .

The Adapazari soils, from which the samples used in this study were taken, are flood plain sediments of fluvial origin. Since the precipitation pattern is the same in such environments all over the world, the results obtained within the scope of the study can also be used in environments with similar formation conditions. When it is desired to be used on soils with different formation conditions, the results should be confirmed by different methods.

The studies that can be done after this study can be summarized as follows:

- The model will decide whether the new data has soil liquefaction potential with a web interface.
- The model can be compared with deep learning methods. Due to the small number of data, this study has not utilized deep learning methods.
- Fuzzy, coarse, and gray clusters can be used for soil liquefaction modeling.

CRedit authorship contribution statement

Mustafa Özsağır: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Caner Erden:** Software, Methodology, Visualization, Data curation, Writing – original draft. **Ertan Bol:** Methodology, Data curation, Writing – review & editing, Investigation, Supervision. **Sedat Sert:** Writing – review & editing, Supervision. **Aşkın Özocak:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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